*Session 4 · Tuesday 8 September 2026 · Items 1 and 2, worked. Items 3 and 4 are PS1; those solutions post to bCourses after the deadline.*

# Answers — items 1 and 2

This is the key to the two items we did **in the room**. Items 3 and 4 of your handout are **PS1 Q4b and Q5**; their worked solutions post to bCourses after the deadline, not here.

If your algebra differs from mine but your answer agrees, you are fine. The prose under each ▶ is the part worth reading — the algebra was never the hard bit.

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## 1 · One site — the whole derivation

$$\mathrm{P}+\mathrm{X} \rightleftharpoons \mathrm{PX}, \qquad K_d = \frac{[\mathrm{P}]\,x}{[\mathrm{PX}]} \implies [\mathrm{PX}] = \frac{[\mathrm{P}]\,x}{K_d}$$

$$f_{\text{bound}} = \frac{[\mathrm{PX}]}{[\mathrm{P}]+[\mathrm{PX}]} = \frac{[\mathrm{P}]x/K_d}{[\mathrm{P}]\,(1+x/K_d)} = \frac{x/K_d}{1+x/K_d} = \frac{x}{K_d+x}$$

A Hill function with $n=1$ and $K=K_d$. The half-point sits at $x=K_d$, which is what a dissociation constant means.

**▶ Why does $[\mathrm{P}]$ cancel, and what would have to be true for it not to?**

$[\mathrm{P}]$ appears in **every term of both the numerator and the denominator**, because every state of the promoter in this model is reached from the empty promoter by multiplying by a factor. The empty state is the reference, and a reference state always divides out of a fraction of states.

It would fail to cancel if the promoter could occupy a state that is **neither empty nor reachable from empty by binding X** — a second conformation, a competing repressor, a state with RNA polymerase bound. Then the denominator carries a term that is not proportional to $[\mathrm{P}]$ and the cancellation breaks. That is the whole subject of session 6: once the denominator is a real partition function over states you chose, "which states did you include" becomes a modelling decision with consequences you can measure.

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## 2 · $n$ sites, all or nothing

$$\mathrm{P} + n\,\mathrm{X} \rightleftharpoons \mathrm{PX}_n, \qquad K_d = \frac{[\mathrm{P}]\,x^{\,n}}{[\mathrm{PX}_n]} \implies [\mathrm{PX}_n] = \frac{[\mathrm{P}]\,x^{\,n}}{K_d}$$

$$f_{\text{bound}} = \frac{[\mathrm{PX}_n]}{[\mathrm{P}] + [\mathrm{PX}_n]} = \frac{x^{\,n}/K_d}{1 + x^{\,n}/K_d} = \frac{x^{\,n}}{K_d + x^{\,n}}$$

To read that as $x^n/(K^n + x^n)$, set $K^n = K_d$:

$$\boxed{K = K_d^{\,1/n}}$$

▶ **Your $K$ is not $K_d$. What is it?**

$K$ is the **concentration of X at half occupancy** — a concentration, in molar. $K_d$ here is an equilibrium constant for a reaction that consumes $n$ molecules of X at once, so it has units of **concentration$^n$**. They are not the same kind of object, and the fastest way to catch the error is dimensional analysis rather than algebra: a quantity with units of $\mathrm{M}^n$ cannot be a half-point.

Two promoters with the **same $K$ and different $K_d$** therefore differ in $n$. They switch at the same input concentration and with different sharpness: same threshold, different steepness. That is exactly the pair of knobs you want separated when you are designing a circuit, and it is why quoting $K_d$ alone tells you almost nothing about how a promoter behaves.

▶ **"No partially bound states" is a lie. Whose lie is it?**

It is the **infinite-cooperativity limit**: binding the first X must make binding the rest so favourable that the intermediate states are never populated. Real cooperativity is finite, so intermediates always have some occupancy and the true curve is shallower than $x^n/(K^n + x^n)$ with $n$ = the number of sites.

It is a *good* lie when the free-energy gain from the first binding event is large compared with $k_B T$ — strongly cooperative systems such as haemoglobin sit close enough to it to be useful, which is where the whole formalism came from.

To find out whether it is a good lie **here**: fit the measured dose–response for $n$ and compare it with the **structurally known site count**. If the fitted $n$ comes back well below the number of sites, the intermediates are populated and the two-state idealisation is doing real damage. That comparison — a fitted exponent against a counted number — is the subject of the last twenty minutes of the session, and of ConcepTest 2.

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**Items 3 and 4** are PS1 Q4b and Q5. Solutions post to bCourses after the deadline. If you are stuck before then, the discussion hour is the place — the answer to Q5 in particular turns on a distinction (*which* fraction you are computing) rather than on algebra, and that is much faster to sort out in conversation.